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Today
Auth vs Authz

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What is the difference between authentication and authorization

The difference between **authentication** and **authorization** is fundamental in security:

🔐 Authentication:

Who are you?
It's the process of **verifying the identity** of a user or system.

- Example: Logging in with a username and password.
- Result: The system knows *who* you are.
- Tools: Passwords, biometrics, tokens, OTPs, etc.

✅ Authorization:

What are you allowed to do?
It's the process of **granting or denying access** to resources based on identity and permissions.

- Example: After logging in, being allowed to access admin pages or edit a document.
- Result: The system knows *what* you can do.
- Tools: Access control lists, roles, permissions.

Feature	Authentication	Authorization
Purpose	Verify identity	Control access to resources
Happens first?	Yes	After authentication
Determines?	Who the user is	What the user can access
Example	Login	Accessing admin dashboard

Would you like an analogy to help visualize the difference?

can you summerize in youtube video, here is the link https://www.youtube.com/watch?v=_zyKvPvh808&t=534s

I can't access or summarize YouTube videos directly from a link unless you provide the content or a transcript. However, if you tell me what the video says about authentication vs authorization (or paste a transcript or key points), I'd be happy to summarize it for you.

Would you like me to help you extract the transcript from that timestamp if you upload it or copy-paste some of it here?

here is the transcript from youtube
[Music]
symmetric encryption and asymmetric encryption we mentioned that asymmetric encryption uses two different keys to do its encryption and decryption
well those two keys are referred to as a key pair and you can use that key pair to do encryption so let me show you how that works
we're going to use pam and jim they're going to use asymmetric keys to securely exchange data with one another
now since these two are two different people they each have their own set of public and private keys
pam has one set of public and private keys she has her public key closer to jim available to jim if he needs it and she has her private key hidden private from anybody else
jim also has his own set of public and private keys
and like we discussed in the last lesson what is encrypted with this public key can only be decrypted with this private key and vice versa
so let me show you how these keys are used to do encryption
let's say jim has a message that he
well what he can do is use pam's public key to encrypt that message
now the only key in the world that can decrypt this back to the original message is the correlating private key which only pam has because again the private key is never shared
that message can be sent across the wire safely knowing that only pam has the required key to decrypt the content of that message and extract the original plaintext
and this is how asymmetric keys are used

to provide confidentiality
recall that confidentiality is the idea
that the data is only accessible to the
intended recipient
and we can use an asymmetric key pair to
ensure data confidentiality
but that's not all you can do with a set
of asymmetric keys
there's actually something else you can
do which is pretty cool
let's say pam now has a message for jim
let's say for this particular message
pam doesn't care about the
confidentiality meaning she doesn't care
who can read this message
but she definitely wants to prove that

it was definitely her that sent this
message
what she can do is use her own private
key to encrypt that message
now the only key in the world that can
decrypt that would be her public key
which jim has and anybody else has
because it's a public key
so this can get sent across the wire
and then jim can decrypt that message
using pam's public key
if jim can decrypt the message
successfully with pam's public key this
actually proves two things
first it proves that pam must have sent
that message this gives us
authentication
if jim was able to decrypt something
with public key it proves that the only
key that could have encrypted it was
pam's private key and pam should be the
only person that has that key therefore
we know pam sent that message
it also proves something else
it also proves that the message was not
modified in transit that is integrity
if someone captured the message in the
middle and changed the content of the
message well when jim tried to decrypt

with a jarbled mess if the decryption
would have worked at all
so if jim was successfully able to
decrypt a message with pam's public key
then we know for a fact the message
hasn't changed since pam encrypted it
with her private key
this process that we just discussed is
what's known as message signatures
you can use an asymmetric key pair to
create a signature which provides
authentication and integrity for what
was signed
so
we just talked through how you can use
an asymmetric key pair to provide both
encryption to get confidentiality and
signatures to get integrity and
authentication
but in reality it's not quite that
simple
remember in the last lesson we talked
through a bunch of limitations that
asymmetric encryption has
namely that it's inefficient to use
asymmetric encryption for bulk data
transfers because asymmetric encryption
is slower and more taxing on the cpu

bulk data what we want to be doing is
using symmetric encryption
but symmetric encryption has this
problem that we have to first establish
a symmetric key on either side of the
wire we have to do it in a secure way
well that symmetric key is a pretty
small piece of data
and recall that we can use asymmetric
encryption for smaller limited data sets
well why don't we use an asymmetric key
pair to share a symmetric key
that is what ssl and tls actually do
let me show you how it works
pam is going to randomly generate a
symmetric secret key this key is just a
string of ones and zeros and pam will
generate them randomly
pain will then take
jim's public key
and encrypt that symmetric key
the only key in the world that can
decrypt this back to the original
symmetric secret key is jim's private
key which only jim has which means pam
is safe to send that across the wire
jim will then take his private key
to decrypt the content of that message
to extract the original symmetric secret
key

and now both parties have established a symmetric key that is identical in a secure way
they can use that symmetric key to protect bulk data
here pam has a message for jim
pam will encrypt that message with her symmetric key
the only key in the world that can decrypt that message is the same symmetric key that was used to encrypt it
that message can be sent across the wire and then jim can use his copy of the symmetric key to decrypt that message and extract the original content
this can be done in either direction for an arbitrary amount of data
here jim is sending something to pam securely using that same symmetric key
now that both parties have the identical symmetric key on either side they can continue to use that for however much data that they want to protect
this concept that we just discussed is referred to as hybrid encryption
hybrid encryption is the idea of using

encryption
now you're using both of those at different points so it's not like you're both symmetrically and asymmetrically encrypting the same content you're using asymmetric encryption to facilitate a key exchange
and then with this key you're then using symmetric encryption to protect bulk data
do you remember those strengths and weaknesses of symmetric and asymmetric encryption we talked about in the last lesson
using hybrid encryption in this way gets you the best of both worlds
you get the strength of asymmetric encryption that you have used the more secure way of sharing a key
and the efficiency of symmetric encryption to protect your bulk data transfer
this method is what every secured communication protocol uses this is how ssl tls protect bulk data transfers this is also how ipsec and ssh protect bulk data transfer
so that is the concept of hybrid encryption

but
what about signatures
we talked about earlier how you can use signatures to attain integrity and authentication on a message by encrypting something with your own private key
but again asymmetric encryption has limitations you can't simply encrypt an entire message with a private key
instead what we'd like to do is simply encrypt a smaller fixed representational sample of that message
if only we had a way of converting messages to smaller representational sizes
well we do we talked about hashing algorithms earlier on in this course
we define them as a thing that specifically exists to take a message of any size and produce a smaller fixed width fingerprint of the original message
and signing is going to use hashing algorithms to simplify the process
let me show you how that's going to work here
is how asymmetric keys are used for generating signatures

she's then going to run that message through a hashing algorithm
that's going to result in a particular digest that digest is then going to be encrypted with pam's private key
the result of that which is the encrypted digest
is the signature that is actually the signature of that message
that gets appended to the message and then both the message and the signature gets sent across the wire
now that signature was created with pam's private key which means on the other side jim is going to use pam's public key to verify the signature
what he's going to do is take the

signature and decrypt it using pam's public key
that's going to result in the digest that pam created
then jim is going to independently calculate a hash of that message
and if the digest that jim got in his calculation matches the digest that pam had sent this proves two things
first it proves that the message has not changed since pam signed it remember

of this message so if anything changed in this message tim would have gotten a different digest
this gives us the property of integrity
the other thing that signatures prove is that only pam could have created the signatures
this signature was created as a result of taking pam's private key and encrypting this digest well if jim was able to decrypt something with paim's public key this proves it was definitely pam's private key that signed it and the only person in the world that has pam's private key is pam
this gives us authentication
so using an asymmetric key pair to generate signatures in this way gives us both integrity and authentication for the content that was signed
now
we've been discussing this from the perspective of a message sent across the wire
but keep in mind many different things can be signed
sure you can sign messages but that's actually pretty rare
often what you see signed is much

single certificate are things like a single piece of software or an email or whatever the case
so this definition of signatures should not be limited to just messages you can sign anything
and that wraps up our discussion of public and private keys
the main takeaway from this lesson is understanding hybrid encryption how we can use asymmetric encryption to securely establish symmetra keys and then we can use the symmetric encryption to protect bulk data at scale
and the other thing to understand is message signatures how they are a function of using your own private key and a hashing algorithm to prove you signed something and it hasn't changed since you signed it
in the next lesson we're going to summarize a lot of what we've discussed so far in this module we're going to tie it all together and show you how ssl actually uses all these different cryptographic tools we discussed
but that's it for this lesson i hope you enjoyed this video i want to thank you

next one

Thanks for sharing the transcript! Here's a concise **summary of the YouTube video** based on it, focused on **authentication**, **authorization**, and related cryptographic concepts:

🔐 Summary of the Video: Public/Private Keys, Encryption & Signatures

🔧 **Asymmetric Encryption:**

- Uses a **key pair**: a **public key** (shared) and a **private key** (kept secret).
- **Confidentiality:**
  - Jim encrypts a message using **Pam's public key**.
  - Only Pam can decrypt it with her **private key**, ensuring **only she can read it**.

✍️ **Message Signatures (Authentication + Integrity):**

- Pam wants to prove *she* sent a message (not necessarily keep it secret).
- She encrypts the **message hash** (digest) with her **private key** — this is the **digital signature**.
- Jim (or anyone) can verify it using Pam's **public key**:
  - **Authentication**: Proves Pam created the signature.
  - **Integrity**: Proves the message hasn't been tam ed with.

- Asymmetric encryption is **slow**, so it's only used to **securely exchange a symmetric key**.
- Once exchanged, both parties use the **symmetric key** for **fast bulk data encryption**.
- This is how protocols like **SSL/TLS, SSH, and IPsec** work.

✅ Key Differences – Authentication vs Authorization:

Concept	Authentication	Authorization
Meaning	Verifying identity (e.g. *Who is Pam?*)	Granting access (e.g. *Is Pam allowed to read this file?*)
Achieved by	Digital signatures, credentials (passwords, certificates)	Access control lists, permissions, roles
In Video?	Yes – via **signature verification** using public keys	No – **authorization** was not the focus of this video

🧠 Main Takeaway:

- **Authentication** and **integrity** are provided using **digital signatures**.
- **Confidentiality** is provided via **encryption**.

data.

Let me know if you'd like a visual diagram or slide version of this summary!

give me a quick explanation of base64 number system and how to convert string from ascii to base64 and what are the equal signs that appear sometimes at the end of base64 strings mean

Sure! Here's a **quick explanation** of **Base64**, how it works, and what the equals signs mean:

🔢 What is Base64?

Base64 is **not a number system** like binary or hexadecimal. It's a **binary-to-text encoding** scheme used to **represent binary data (like images or files)** in a text format — especially useful for sending data over media designed to handle text (like email or JSON).

📦 How Base64 Works:

- Splits those 24 bits into **4 groups of 6 bits**.
- Maps each 6-bit group to a character from the **Base64 alphabet** (64 characters):

```
css                                                    Copy  Edit

A–Z, a–z, 0–9, +, /
```

So, every 3 ASCII characters (3 bytes = 24 bits) become 4 Base64 characters.

🔄 Convert ASCII to Base64:

Example with the string `"Man"`:

1. ASCII of "Man" → bytes:

```
ini                                                    Copy  Edit

M = 77 → 01001101
a = 97 → 01100001
n = 110 → 01101110
```

2. Concatenate bits:
   `01001101 01100001 01101110` → 24 bits
3. Split into 6-bit chunks:
   `010011 010110 000101 101110`

   `19 22 5 46`
5. Map to Base64 chars:
   `T W F u`
   → So, `"Man"` → `"TWFu"`

❓ What do the `=` signs mean?

- Base64 must encode in groups of **4 characters**.
- If your input isn't a multiple of 3 bytes (e.g., only 1 or 2 bytes), **padding is added** with `=` to complete the set of 4 Base64 characters.

Input bytes	Base64 output length	Padding
3 bytes	4 chars	0 `=`
2 bytes	4 chars	1 `=`
1 byte	4 chars	2 `=`

Examples:

- `"Ma"` → `"TWE="`
- `"M"` → `"TQ=="`

📌 Summary:

- Base64 encodes binary data into 64-character alphabet text.
- 3 bytes → 4 Base64 characters.
- `=` signs are **padding** to align to multiples of 4 characters.

Would you like a short code example to do this conversion in Python or JavaScript?

I will tell you what i know about json web token and you correct me if I have some mistake
first JWT make of three part, they are header payload and signature
the header and the payload are simply json objects combined with a private key
they get hashed and then converted to base64 and that's how the server creates the signature
and it also converts the header and the payload to base64 and separate them by full stops and then hand it the the user

when verifying
the server recalculate a new signature with the provide token from the user
of course using the private key only the server has
if the newly calculated signature matches the provided signature on the user token then everything is good and the user is allowed to access what ever th asked for

do I have any mistakes? Especially in the hashing part

You're **very close** and mostly correct — great job! 👏
I'll clarify and correct a few small points, especially regarding **hashing**, **signing**, and **the role of the private key**.

✅ Correct Points:

- A **JWT** (JSON Web Token) has **3 parts**:
  `header.payload.signature` — separated by dots (`.`)
- **Header** and **Payload** are **JSON objects**, **Base64URL-encoded**.
- The **Signature** is used to verify the **integrity** and **authenticity** of the token.
- The server **verifies** the signature to allow or deny access.

⚠️ Corrections and Clarifications:

🔸 **1. "Combined with a private key" is misleading**

> You said: *"header and payload are simply JSON objects combined with a private key..."*

✅ Actually:

- The **header and payload are NOT hashed with a private key**.
- Instead, they are **Base64URL-encoded**, **concatenated** with a `.` (dot), and **signed** with a secret or private key, depending on the algorithm.

🔸 **2. It's not just hashing — it's signing**

> You said: *"they get hashed and then converted to base64..."*

🔁 Actually:

- The **signature is created** by applying a **digital signature algorithm**, not just a hash.
- Example (with HMAC SHA256):

```plaintext
signature = HMAC-SHA256(base64url(header) + "." + base64url(payload), secret)
```

  - This is a **message authentication code**, not just a hash.
  - With RSA or ECDSA, a **private key** is used to create a digital signature.

🔐 Summary:

Part	Role
Header	Contains the algorithm (e.g., HS256, RS256), type
Signature	Verifies integrity & authenticity of header + payload

- ✅ Server uses **secret (HS256)** or **private key (RS256)** to sign the token.
- ✅ On verification, the server **recomputes the signature** using the **same algorithm** and compares it to the one in the token.

✅ Final Verdict on Your Description:

You got the **structure, flow, and purpose** right — just needed precision around the **signing process** (not "hashing with private key"). JWT signing involves more than just hashing — it's about **cryptographic signing** with a key.

Would you like a diagram or code snippet to illustrate how this works with HMAC or RSA?

so the signature is the result of applying fancy hashing and encryption algorithms on the base64 of the header and the payload and the private key after appending them together

You're **very close again**, but let me fine-tune the wording — this is important for clarity:

✅ Corrected Explanation:

> **The signature in a JWT is the result of applying a digital signature algorithm (e.g., HMAC or RSA) to the base64url-encoded header and payload, using a secret or private key.**

It's not just a "fancy hash + encryption" — it's **digital signing**, which uses hashing **as part of** the signing process.

🔧 What Actually Happens:

1. **Base64URL-encode** the header and payload:

```ini
encodedHeader = base64url(header)
encodedPayload = base64url(payload)
```

2. **Concatenate** them:

```ini
message = encodedHeader + "." + encodedPayload
```

3. **Sign the message** using:
   - `HMAC` with a **secret key** (symmetric, e.g., HS256), or
   - `RSA` / `ECDSA` with a **private key** (asymmetric, e.g., RS256)

   Example:

```js
signature = HMAC_SHA256(message, secret)     // for HS256
```

   or

```js
signature = RSA_sign(SHA256(message), privateKey)  // for RS256
```

4. The signature is also Base64URL-encoded and added:

```ini
JWT = encodedHeader + "." + encodedPayload + "." + encodedSignature
```

❗ Important Distinction:

- **HMAC**: Shared **secret key** used for both signing and verification (HS256).
- **RSA/ECDSA**: Server signs with **private key**, and others can verify using the **public key** (RS256, ES256).

Yes, the signature is the result of applying a secure **hash + key-based signing algorithm** to the base64-encoded header and payload — **but it's not encryption**, and the key isn't just "appended." It's used **inside** the signing algorithm (like HMAC or RSA).

Want a quick visual diagram to help cement this?

correct me if I'm wrong
Sign the message means that you take the message and hash it and then encrypt the result with the private key

You're **mostly right**, and that's a very common mental model — especially for **RSA signatures** — but let's refine it with precision:

✅ Your Understanding (Rephrased):

> *"Signing a message means hashing it and then encrypting the hash with the private key."*

✅ That is **correct for RSA-style signing, conceptually.**

But depending on the algorithm (e.g., RSA, ECDSA, or HMAC), the signing process is **not always literally "encrypt the hash with the private key."**

Let me break it down by algorithm:

🔐 RSA (e.g., RS256)

Yes, signing ≈

1. **Hash** the message (e.g., with SHA-256)
2. **Encrypt the hash** using the **private key**
   → That encrypted hash is the **signature**

This is the classic mental model of a **digital signature**.

🔐 ECDSA (Elliptic Curve Digital Signature Algorithm)

ECDSA **does not encrypt the hash**.

Instead, it uses:

- A cryptographic **hash function** (e.g., SHA-256), and
- Elliptic curve math to generate a **signature pair (r, s)** using the private key

It's more complex than RSA, but the key idea remains:

- Sign = prove you hold the private key for that hashed message.

HMAC is not encryption at all — it's a **symmetric key hashing algorithm**.

- You combine a **shared secret key** with the message and hash them together.
- It produces a signature-like value, but **no private/public keys** involved.

📌 Summary:

Algorithm	What "signing" means
RSA	`signature = encrypt(hash(message), privateKey)` ✅ your description applies
ECDSA	Uses curve math on the hash with the private key — not encryption
HMAC	Uses secret + hash; no encryption, no keys to encrypt with

---

So:
✅ Your definition **works well for RSA**, but
❌ it's not how all signing algorithms work under the hood — especially **HMAC and ECDSA**.

Would you like a visual chart comparing these three signing methods?

You combine a shared secret key with the message and hash them together.
isn't that what I said above when I said the private key and the header and the payload get combined and hashed

---

Yes — you're *very close* again! Your earlier statement:

> "the private key and the header and the payload get combined and hashed"

...is **almost exactly what HMAC does**, *but* let's zoom in on two key differences:

🔍 1. Terminology: "private key" vs "shared secret"

- In **HMAC (e.g., HS256)**, the key is a **shared secret**, *not* a private key in the public/private sense.
- A **private key** usually implies a corresponding **public key** (used in **asymmetric** cryptography like RSA/ECDSA).
- But HMAC is **symmetric**, so the same **secret key** is used by both the sender and the verifier.

> ✅ So your idea is right in **HMAC logic**, just the term "**private key**" is technically not accurate in this case.

🔍 2. How they are "combined"

HMAC doesn't just "append" or "concatenate" the key and the message. It uses a specific internal structure:

```vbnet
HMAC(key, message) = hash((key ⊕ opad) || hash((key ⊕ ipad) || message))
```

Where `⊕` is XOR, and `ipad/opad` are constant byte patterns.

> ❗ So it's **not a simple hash(key + message)** — it's a more secure construction that prevents certain attacks.

✅ In simple terms though:

✅ **In simple terms though:**

You're exactly on the right track conceptually:

- **Yes**: you hash the message using a **secret key**, and
- **Yes**: this produces a signature to prove the sender had that key.

Would you like a concrete example of how an actual JWT HMAC signature is built, step by step?

how it's not private key
in the course I learn Node.js, the instructor said that I should create a strong key and keep it private in a .env file because using this key the
so if someone else have that key, he could create unauthorized tokens and my server won't be able to detect them

You're absolutely right to question that — and your reasoning is **completely valid**.

The key you're describing **is private** in practice — meaning it should be **kept secret** — but the confusion arises because cryptography uses the word **"private key"** in a very specific way.

🧠 Let's clarify the terminology:

✅ **In your Node.js course (HS256, HMAC):**

- You generate a strong secret key (e.g. `JWT_SECRET`).
- You keep it in a `.env` file.
- Only the server knows it.
- It is used to **sign** and **verify** JWTs.

> ✅ This is a **private secret** in a practical sense — you're right!

But...

❗ **In cryptographic terminology:**

- A **private key** implies there is a **public key** — that's **asymmetric cryptography** (e.g., RS256, ECDSA).
- A **shared secret** (like in **HS256**) is part of **symmetric cryptography**:
  - The same key is used for both **signing** and **verifying**.
  - If anyone else gets the key, they can both forge and verify tokens.

So technically:

- ✓ You're **keeping the key private** ✅
- ✗ But it's **not a "private key"** in the cryptographic sense, because there's no **public counterpart**

🔧 TL;DR:

Concept	What you said	Cryptographic term
A secret key that only the server knows	"Private key"	Shared secret key (HMAC, symmetric)
A key used with a public key to sign	✗	Private key (asymmetric, e.g., RSA)

✅ You're right in *practice*, just slightly off in *terminology* — and now you know why.

Would you like a comparison of HS256 (HMAC) vs RS256 (RSA) JWT tokens?

so I should have said private secret since it's not public and there is not counterpart to it

Exactly — well said!

✅ The best term is **"secret key"** or **"shared secret"**, and yes, it's private because:

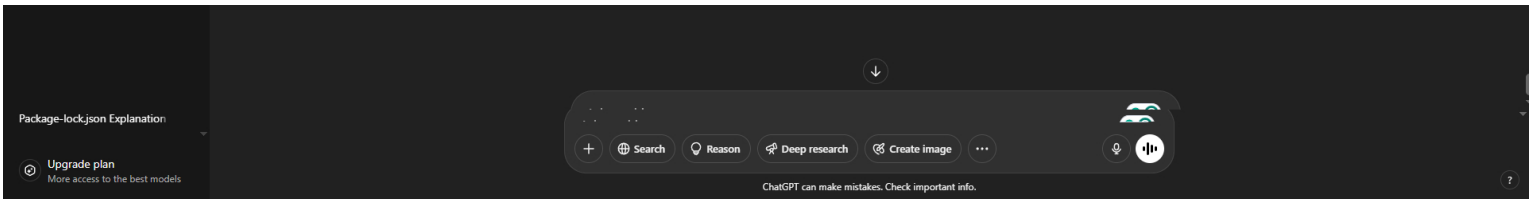
- It's **never shared publicly**.
- It has **no public counterpart** like asymmetric keys do.
- If someone else gets it, they can **both sign and verify** tokens — so it must stay secret.

So in plain language, you could say:

> "The server uses a **private secret key** (or just **secret**) to sign the JWTs with HMAC. It must never be exposed."

That keeps things both accurate **and** understandable.

Would you like a quick visual or example to compare HMAC (HS256) vs RSA (RS256) JWTs?


Package-lock.json Explanation
Upgrade plan
More access to the best models
Search
Reason
Deep research
Create image
ChatGPT can make mistakes. Check important info.